

Beaming AR: A Compact Environment-Based Display System for Battery-Free Augmented Reality

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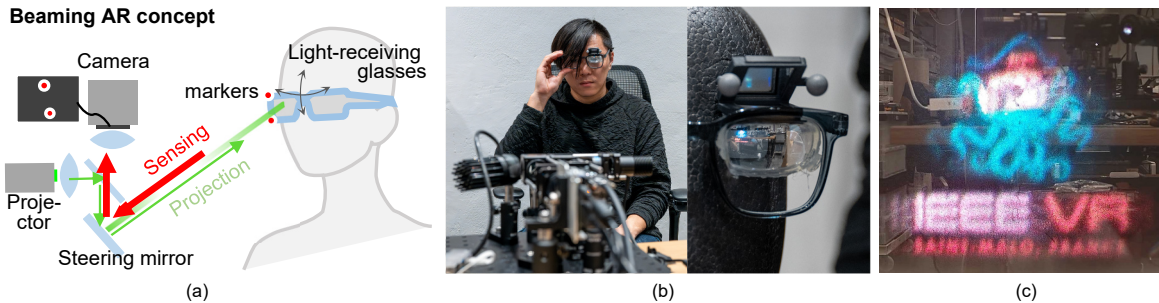


Figure 1: Overview of the Beaming AR demonstration: (a) Core system architecture illustrating the co-axial projection and sensing paths through a single steering mirror. The environment-mounted unit contains all active components while users wear only passive light-receiving glasses with retroreflective markers. (b) Left: Photos showing a user wearing the modified glasses and the bench-top projection-tracking unit. Right: A close-up view of the light-receiving glasses with retroreflective markers. Visible light projected from the system is captured on the glasses' light-receiving optics. (c) Example of projected AR content as seen through the light-receiving glasses, showing floating multi-color graphics overlaid on the real-world view.

ABSTRACT

Beaming AR demonstrates a new approach to augmented reality (AR) that fundamentally rethinks the conventional all-in-one head-mounted display paradigm. Instead of integrating power-hungry components into headwear, our system relocates projectors, processors, and power sources to a compact environment-mounted unit, allowing users to wear only lightweight, battery-free light-receiving glasses with retroreflective markers. Our demonstration features a bench-top projection-tracking setup combining steerable laser projection and co-axial infrared tracking. Conference attendees can experience this technology firsthand through a receiving glasses, demonstrating how environmental hardware offloading could lead to more practical and comfortable AR displays.

Index Terms: Beaming display, augmented reality, near-eye display, steering projector.

1 INTRODUCTION

Augmented Reality (AR) overlays virtual content onto the physical world, enabling interactive experiences across diverse applications. However, conventional head-mounted displays (HMDs) remain heavy and cumbersome, primarily due to their integrated batteries, processors, and optical components. These limitations cause user discomfort, excessive heat generation, and brief operating times, hampering widespread adoption [3].

To overcome these challenges, we have developed the **Beaming Display** concept [5]. This approach redistributes system hardware between the user and environment, departing from traditional all-in-one wearable designs. Instead of integrating all components into the headset, Beaming Display houses power, computing, and projection elements in an environment-mounted steering projection-tracking

unit. Users wear only passive, lightweight eyewear that receives and directs projected light to their eyes.

While previous works explored various aspects of the Beaming Display concept, each prototype focused on specific technical challenges. The initial demonstration proved basic feasibility with AR markers but lacked practical considerations [5]. Later iterations addressed individual aspects: one achieved low latency using active LED markers on a headset [4], another expanded tracking volume with external motion capture [2], and the others developed slim optical designs for the receiving glasses [1, 6]. However, none of these prototypes achieved a complete, integrated system suitable for practical demonstration.

This demonstration bridges that gap, offering hands-on experience with environmental compute offloading and passive AR eyewear (Fig. 1(a)). We present a refined minimum viable product (MVP): a compact steering projector paired with a passive, marker-equipped headset in an integrated bench-top setup. This configuration lets attendees experience battery-free AR firsthand while showcasing practical improvements in calibration and form factor that make environment-based AR both feasible and compelling.

2 IMPLEMENTATION

The core projection-tracking unit is mounted on a 300mm x 300mm optical board, integrating three primary components: a visible-light laser projector, a near-infrared (NIR) LED source, and a CMOS sensor camera (Fig. 2 left). Two mirrors manage the optical paths: a cold mirror separates visible and NIR wavelengths, while an NIR half mirror splits the camera's observation path from the IR illumination path. An Optotune steering mirror (MR-15-30-PS-25x25D) with a $\phi 25$ mm clear aperture controls the projection direction.

2.1 Projection-Tracking Unit

Our core unit integrates three optical paths: projection, illumination, and imaging into a bench-top optical board (Fig. 2).

Projection Path The visible display beam originates from a commodity laser projector (Ultimems HD301D1, 1280x720) (Fig. 2 right. Blue arrows). While projectors typically offer wide projection

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angles suitable for standard display uses, our application requires a narrower beam to achieve precise steering projection on an eyewear. We thus developed a custom lens assembly that reduces the projection angle while maintaining the beam's focus distance. This modified beam then reaches the steering mirror for projection onto the user's eyewear.

NIR Source Path For marker illumination, we employ an IR LED (Thorlabs M850L3 LED, 850nm) (Fig. 2, orange path). The NIR light is collimated by an $f=45\text{mm}$ achromatic lens before following a path through both the NIR half mirror and cold mirror. The beam then reflects off the steering mirror to illuminate the retroreflective markers on the user's eyewear.

Camera Path The imaging system features a pupil-transfer optical design built around a C-mount CMOS camera sensor (Ximea MQ013RG-ON, 1280×1024 , 60Hz) (Fig. 2, red path). A relay lens system—using two $f=75\text{mm}$ achromatic lenses paired with a C-mount lens (RICOH FL-CC7528-2M, $f=75\text{mm}$)—creates a conjugate relationship between the sensor's entrance pupil and the steering mirror. This arrangement achieves two critical goals: it prevents the camera's field of view from being clipped by the steering mirror while expanding the effective tracking field of view. Marker reflections travel back through the steering and cold mirrors, then enter the camera system via a half mirror for real-time marker detection.

2.2 Light-receiving Glasses

The light-receiving glasses integrate three essential optical elements: a light-receiving screen, a collimating lens system, and a waveguide for see-through overlay of projected content. Our current prototype repurposes optical components from a commercial bird-bath-type AR display (Nreal Air), integrating a screen path to accept externally projected light. Specifically, we replace the built-in microdisplay module with a mirror assembly and diffusion film that redirects and conditions the incoming projected beam (Fig. 1(b) right). The glasses frame is equipped with two retroreflective markers for robust position tracking. While this implementation can demonstrate the core concept, future iterations would benefit from a custom waveguide-based light-receiving system optimized for our specific optical requirements [6] and 6DoF tracking with more markers [7]

2.3 Operation and Tracking

During operation, the Optotune mirror directs both visible and NIR beams toward the user's eyewear. Our tracking algorithm uses OpenCV's blob detection to locate two retroreflective markers on the glasses and continuously adjusts the steering mirror using PID control to maintain the markers' midpoint at the camera's center. The system operates in a single-threaded processing loop—capturing camera frames, detecting markers, and issuing mirror control commands sequentially—with timing constrained by the camera's frame rate of 60 Hz (16.6 ms per frame). While the steering mirror itself offers a rapid response time of 100 microseconds, the overall system latency is primarily determined by this sequential processing architecture.

Although our demonstration maintains AR content synchronization with head movements sufficient for demo purposes, the significant gap between camera and processing latencies versus mirror response time suggests that the system's performance could be enhanced through both higher-speed cameras and parallel processing implementations.

3 DEMO PLAN

We will present our bench-top steering projector-tracking setup in an open booth space, allowing participants to directly observe and try on the passive eyewear (Fig. 1(b)). Attendees will wear light-receiving glasses with built-in retroreflective markers. As they move within a designated viewing area (approx. 1-2m from

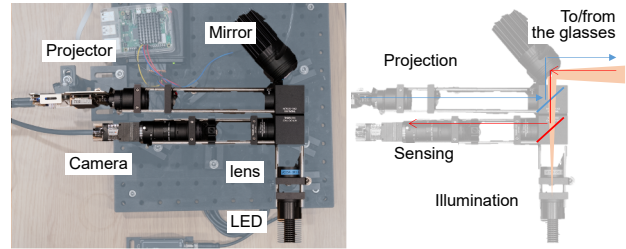


Figure 2: Implementation of the integrated projection-tracking unit: (left) Top-down view of the optical assembly showing key components—laser projector, camera, NIR LED, lens system, and steering mirror—mounted on a 300mm $\times$ 300mm optical board. (right) Schematic diagram illustrating the co-axial optical paths for projection, sensing, and illumination through the shared steering mirror.

the optical board), the camera tracks their head position in real-time. Floating icons or simple text overlays appear aligned with the user's viewpoint (Fig. 1(c)). By turning their head or shifting their body, participants see the projected graphics adapt to their motion. All power, computing, and projection elements remain on the optical board, keeping the eyewear lightweight and free of cables or batteries.

4 CONCLUSION

We presented a practical implementation of the Beaming Display concept, integrating projection, tracking, and control into a compact bench-top system. By combining co-axial laser projection and marker tracking with lightweight receiving glasses, we demonstrate how environmental hardware offloading can enable comfortable AR experiences without on-glasses batteries or processors. Through this hands-on demonstration, we aim to showcase the potential of environment-based AR architectures and inspire further research toward everyday AR displays that elegantly balance wearability and functionality.

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